

PRODUCT GUIDE: PANTOGRAPH & PENDULUM WIPERS

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KARACA
SHIP WINDOWS & WIPERS

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GENERAL INFORMATION AND SECURITY SUMMARY

Safety Precautions

WARNING!

BE AWARE OF POTENTIAL INJURIES!

BE SURE TO READ AND FOLLOW THE INSTRUCTIONS BELOW BEFORE WORKING ON THE WIPER SYSTEM!

If the complete windshield wiper systems are to be installed by the customer, we do not have any right to supervise or intervene in the installation process. Therefore, we cannot be held responsible for any issues arising from installation errors. In cases where installation or maintenance instructions are insufficient or if any specific problems occur, please contact KARACA GEMİ Customer Service directly.

Most wiper motors feature a park setting that allows the motor to move to the park position through the vehicle's electrical system, even when the wipers are in the "off" position. For this reason, **the wiper arm should not be installed during this time, and no one should have their hands, fingers, or any body parts near the system while the wiper system is operating.**

Even small wiper motors cannot be stopped or braked by hand.

Never approach the area where the drive belt is located while the system is operating!

When connecting the wiper motor to the ship's electrical system, falling objects may cause injuries even if the wiper switch is in the off position. Therefore, **objects such as screwdrivers must not be left in the wiper system installation area.**

Ensure that the equipment is handled carefully. **Do not drop or strike** the equipment against hard surfaces. Do not apply any impact to the equipment during installation; otherwise, the wiper system may become mechanically deformed, which can lead to premature failure of the wiper mechanism.

1. INTRODUCTION

This user manual has been prepared to introduce the glass wiper mechanism used on ships and to ensure its proper use. The mechanism in question is a specialized system developed for cleaning ship windows and plays a critical role in providing a safe, efficient, and uninterrupted field of vision in the maritime industry.

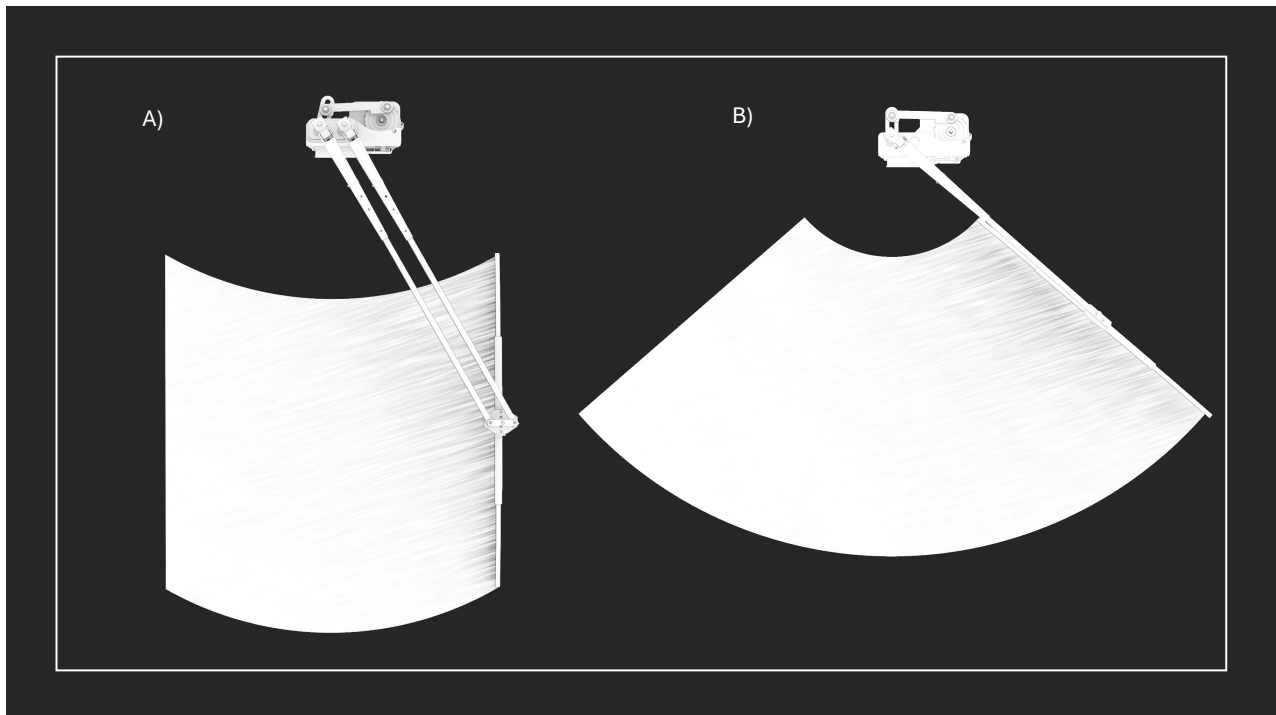


Figure 1. A) Pantograph B) Pendulum

The ship window wiper mechanism is designed to clean all glass surfaces on a vessel, primarily to enhance crew safety during sea voyages and under adverse weather conditions. Rain, seawater, and other contaminants can obstruct visibility on ship windows, making navigation more difficult. These wiper systems quickly remove water accumulation and dirt from the glass surfaces, helping the crew maintain a clear and uninterrupted view during navigation.

The wiper sweep area is determined based on the standing position on the vessel, the user's height, and the field of view. This approach ensures that the areas requiring cleaning are clearly defined and provides a safe voyage by maintaining the operator's field of vision at an optimal level.

2. WIPER DESCRIPTION AND FEATURES

Pantograph and Pendulum wipers are Heavy-Duty wipers equipped with an internally mounted electric motor. The wipers can be installed at the top, bottom, right, or left side of the window. The mechanical specifications are provided in Table 1.

The main components of the windshield wiper system are:

1. Wiper casing
2. Wiper motor
3. Wiper arm
4. Wiper blade

Mechanical Features	
Wiper Mechanism Dimensions	140x145x250
Mechanism Weight	3 Kg
Material	S235JR
Motor Operating Temperature Range	-30°C to +70°C

Tablo 1. Wiper Mechanisö Mechanical Features

2.1. WIPER MECHANISM

The length of the wiper mechanism is determined based on the pressing distance to the glass (Figure 2). The minimum motor shaft length is 70 mm. This value may vary depending on the required contact force and distance of the wiper to the glass. In pantograph and pendulum wiper systems, the maximum sweep angle is 100 degrees. By increasing the number of blades on the arm, maximum length can be achieved. This configuration ensures flexibility and adaptability of the system by meeting the requirements of different stroke lengths.

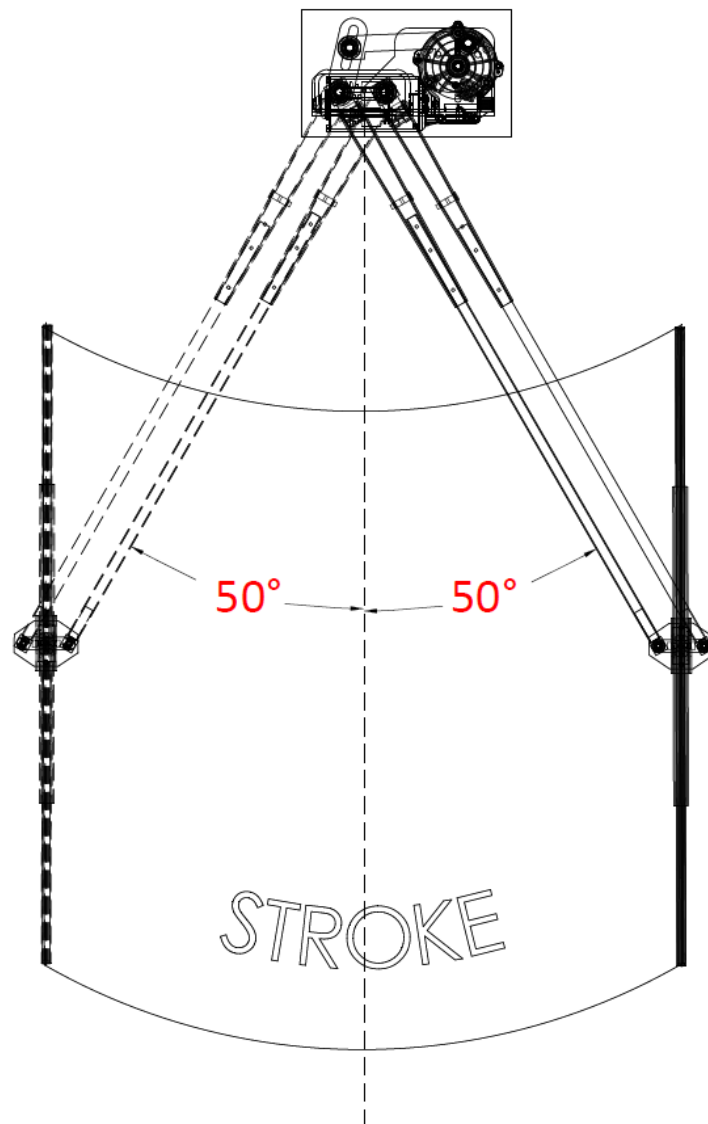
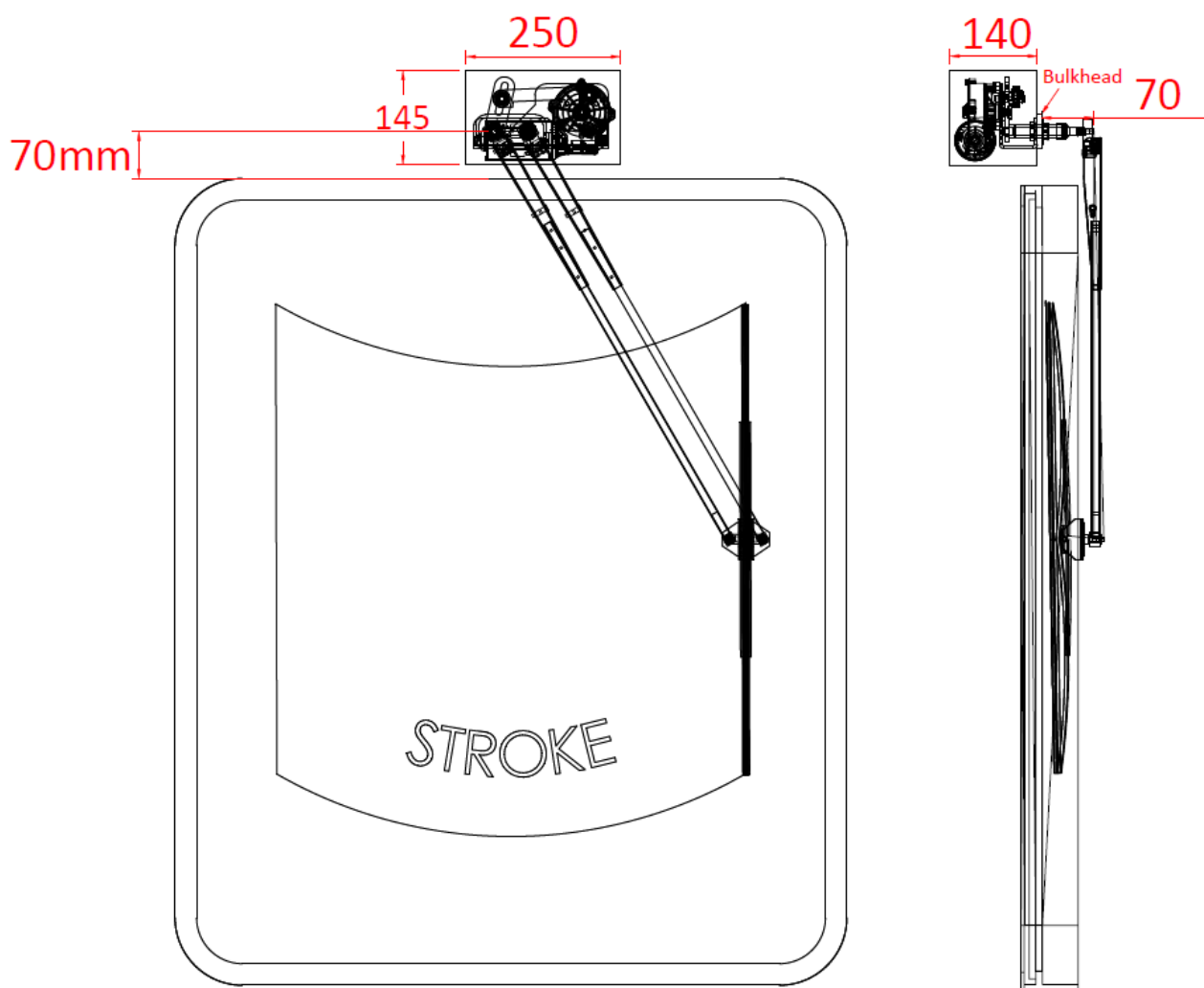


Figure 2. Maximum Wiper Arm Angle



Wiper systems are designed and manufactured to suit different types of glass (Figure 4). Wipers compatible with flat, curved, or specially shaped glass surfaces provide effective cleaning performance on all surfaces. This variety meets the varying needs depending on the type of vessel and operating conditions, enhancing both functionality and durability. Selecting the correct wiper is critically important to ensure a clear field of vision and to prevent damage to the glass surface.

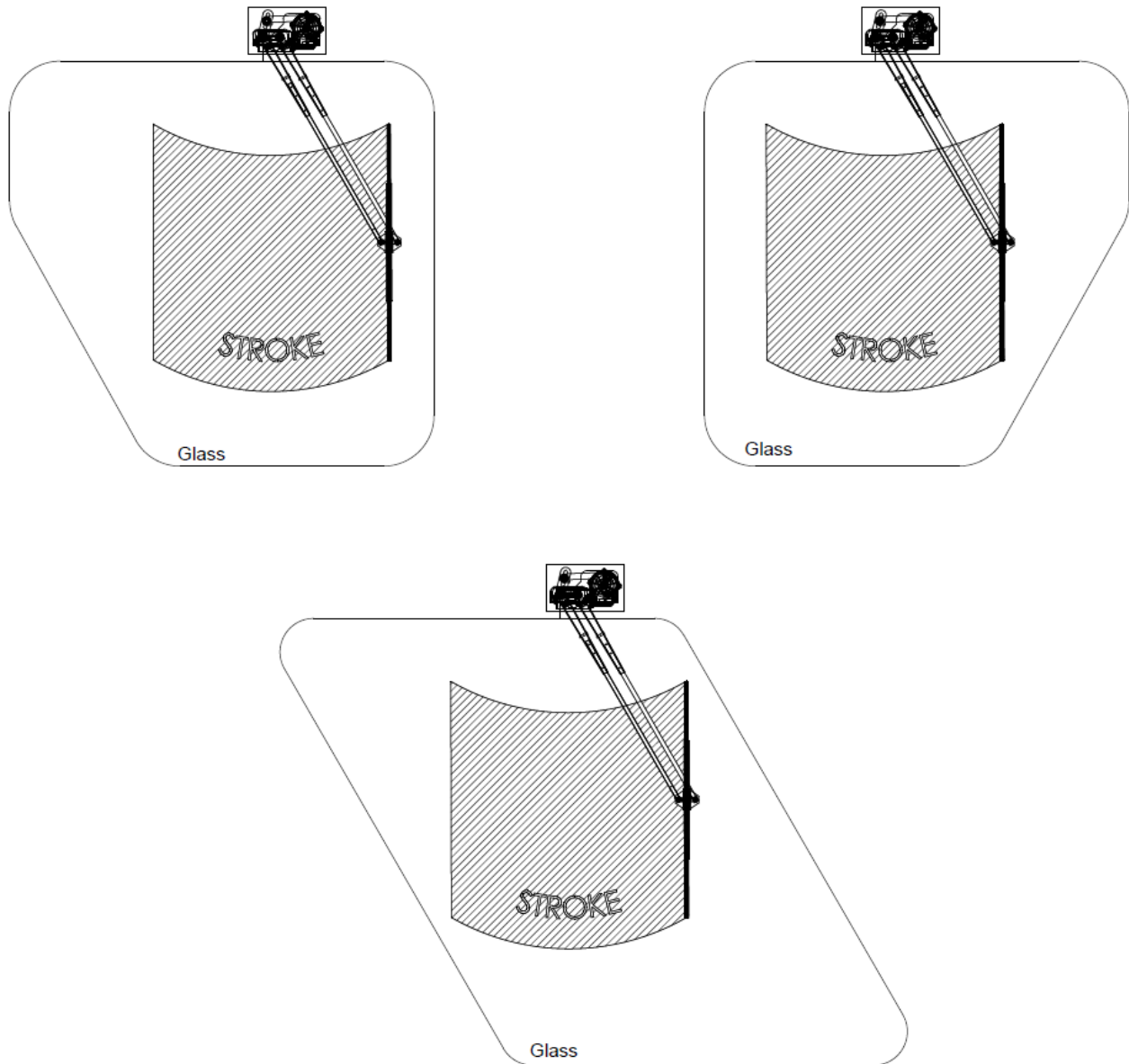


Figure 4. Wiper Types According to Different Glass Types

Pantograph and pendulum wipers operate with a 24V DC motor. The motor is positioned at the center, right, or left side of the glass to maximize the wiping area. The park position can be set to the right or left side depending on the glass structure and user preference. All electric motors include a worm reduction gearbox. The windings have Class F insulation. (Figure 4)

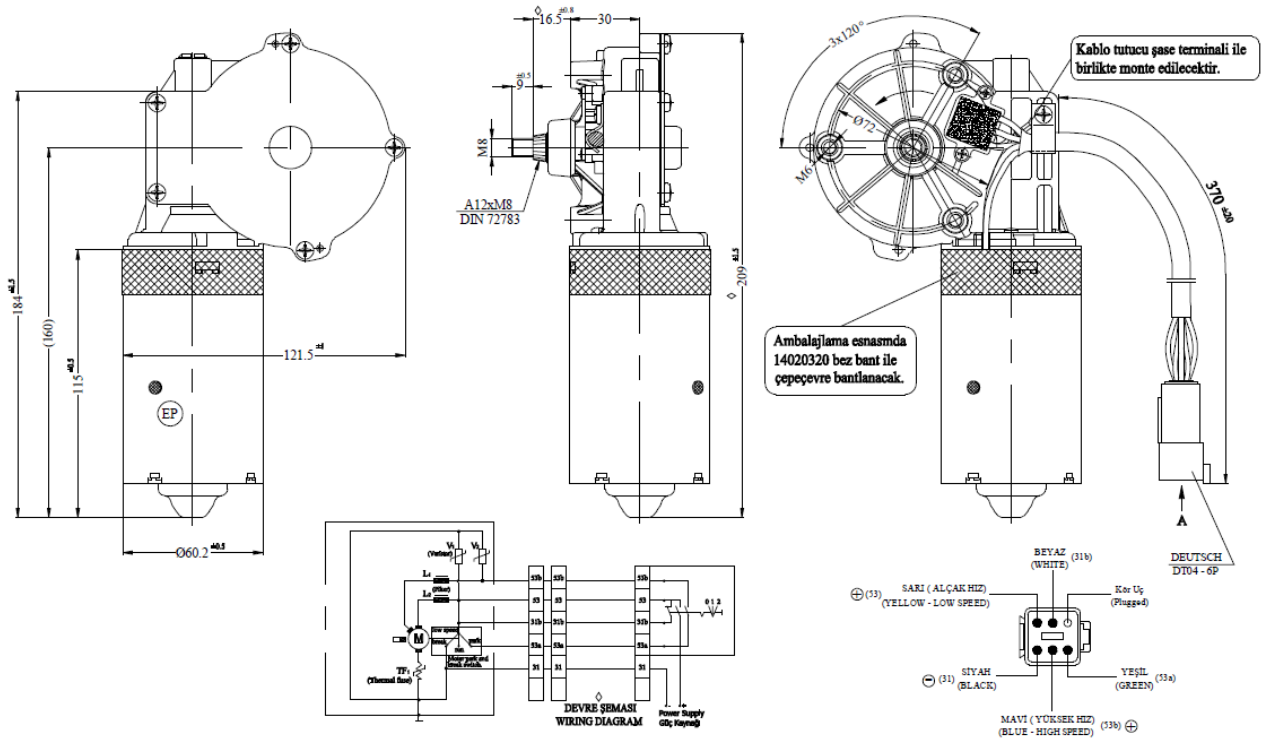


Figure 5. Engine of Pantograf & Pendulum

U (V)	I (A)	n (rpm)	T (N.m)	Titreşim Vibration	Breakaway Torque Koparma Torku	Breakaway Current Koparma Torku
24V	Max. 1	26 ±4	No load	Max. 2.5 g	≥ 40 Nm	≤ 20A
	Max. 1.5	43 ±4	Boşta			
	1.8 ±0.3	24 ±4	6			
	2.4 ±0.3	39 ±4	3			

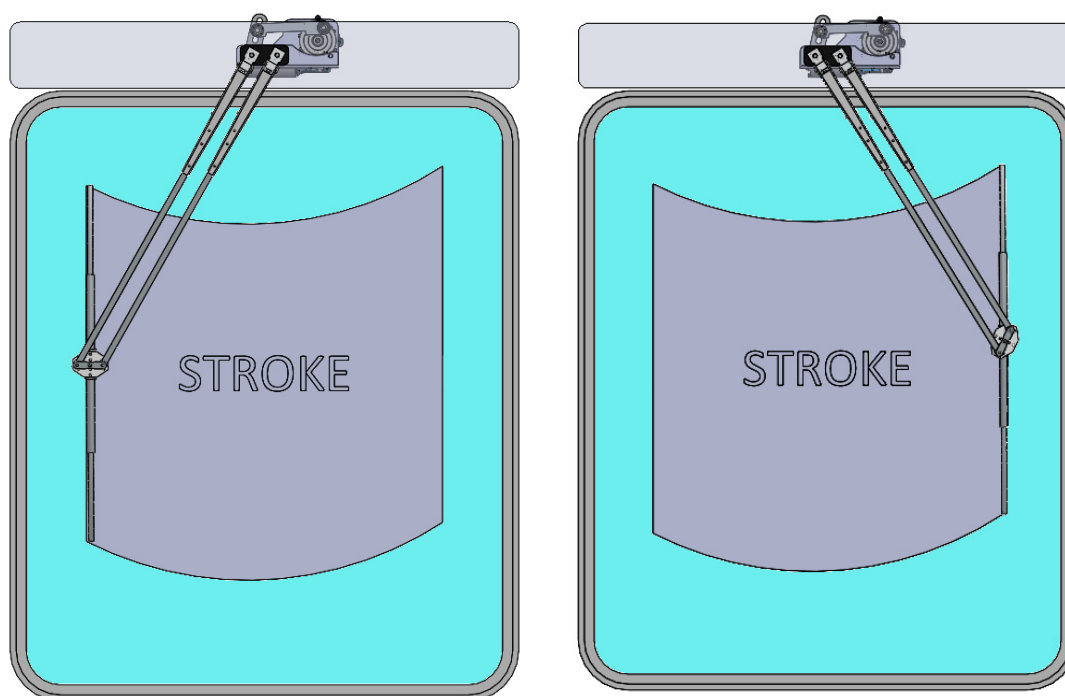


Figure 6. Motor Location Options

The motor's technical dimensions are shown in Figure 5, with the motor occupying a volume of 140 x 145 x 250 mm.

The motors feature a control board that allows for two-speed adjustment, with speeds of 360 rpm and 280 rpm respectively (Table 2).

	Control Unit	
Step	1	2
Rpm	360	280
1 m/sec	0,85	1,1

Tablo 2. Motor Control Board Data

A 24 VDC motor normally draws 2 amps under regular operating conditions, but this current can rise up to 8 amps under stall or heavy load situations. To protect the motor in such cases, it is recommended to use a 10-amp fuse (Table 2). These fuses do not blow during normal operation; however, if the wiper system jams, the fuse will activate to prevent motor damage. It is advised to install a separate fuse for each wiper. Fuses are not supplied by KARACA Ship

and must be installed by the ship. New control boards include built-in fuse protection circuits, which enhance system safety. Motor specifications are provided in Table 3.

Nominal Voltage	Full load current at 50/60Hz	Fusing Value 50/60 Hz	Speed (m/sn)
24 VDC	2 Amper	10 Amper	0,85

Tablo 3. Motor Features

2.2. WIPER ARM

Wiper arm lengths range from a minimum of 300 mm to a maximum of 750 mm. This wide range offers flexible usage for different types of vessels and wiper systems. Selecting the correct arm length plays a critical role in ensuring an effective wiping area. When choosing the wiper arm, it is essential that in both pantograph and pendulum models, the arm does not collide with the frame during the wiping motion. This consideration is carefully taken into account during selection.

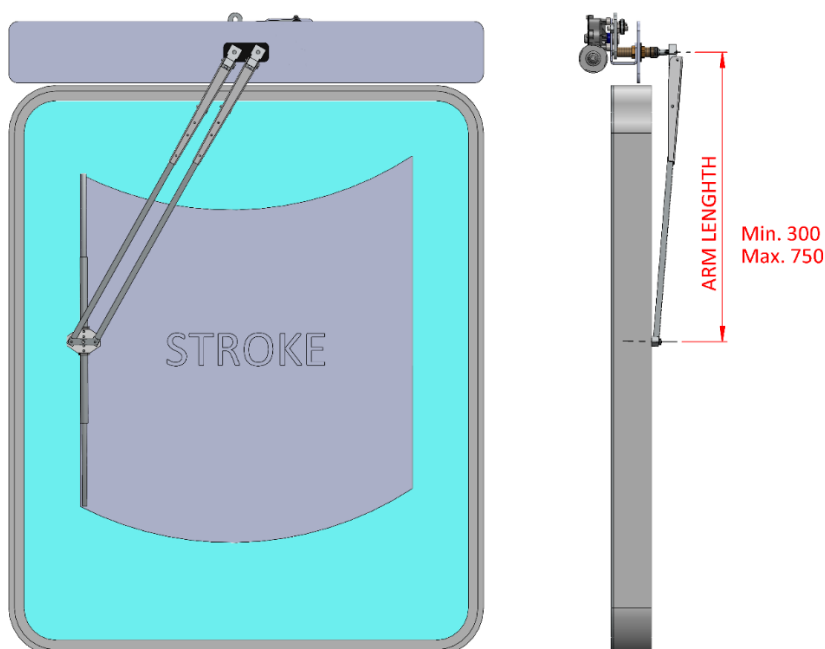


Figure 7. Min - Max Arm Length

The selection of maximum and minimum arm lengths varies depending on the frame structure and the position of the wiper motor mechanism. This standard aims to enhance the system's functionality and reliability, ensuring optimal cleaning performance for windows of different sizes.

When the angle between the wiper arm and the zamak mounting part (the X angle in Figure 8) exceeds 20 degrees, the wiper arm must be bent. This adjustment is made by bending a 12x12 mm flat bar. These design details are critically important for both functionality and safety.

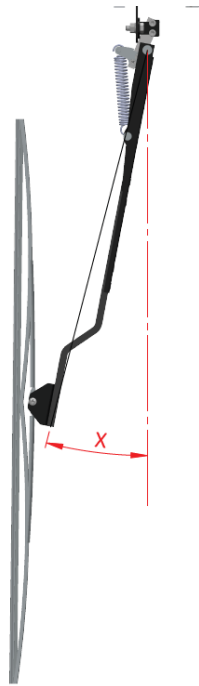


Figure 8. Wiper Arm Bending

2.3. WIPER BLADE

The wiper blade length ranges from a minimum of 300 mm to a maximum of 1200 mm. These dimensions are optimized to provide compatible wiping performance for different window frames. Selecting the correct blade length increases the efficiency of the cleaning area.

The blades are designed with an overlapping double-blade structure. They provide a 31-degree operating angle through slots in the mounting plate (Figure 9). This design maximizes the movement range of the wipers, offering a wide field of vision.

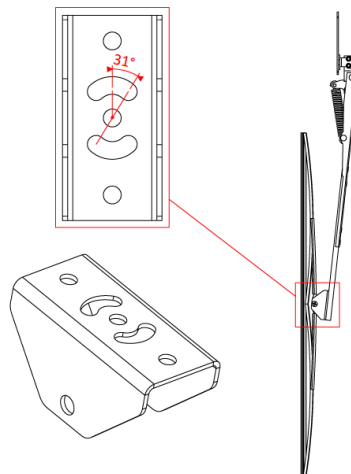


Figure 9. Maximum Wiper Blade Angle

The springs integrated into the wiper blades play a crucial role in applying optimal pressure onto the glass surface, enhancing cleaning efficiency. This design ensures tighter contact between the blades and the glass, contributing to effective removal of water, dirt, and other debris.

3. AUXILIARY COMPONENTS

Auxiliary elements are used to enhance the efficiency of wiper mechanisms. Spray nozzles fall into this category.

3.1. SPRAY NOZZLES

The clean water source supplies the spray function through an 8 mm compression fitting for the hose (Figure 10). The spray nozzles are specially adjusted for each window. For the wiper blades to perform effectively, proper positioning of the nozzles is essential.

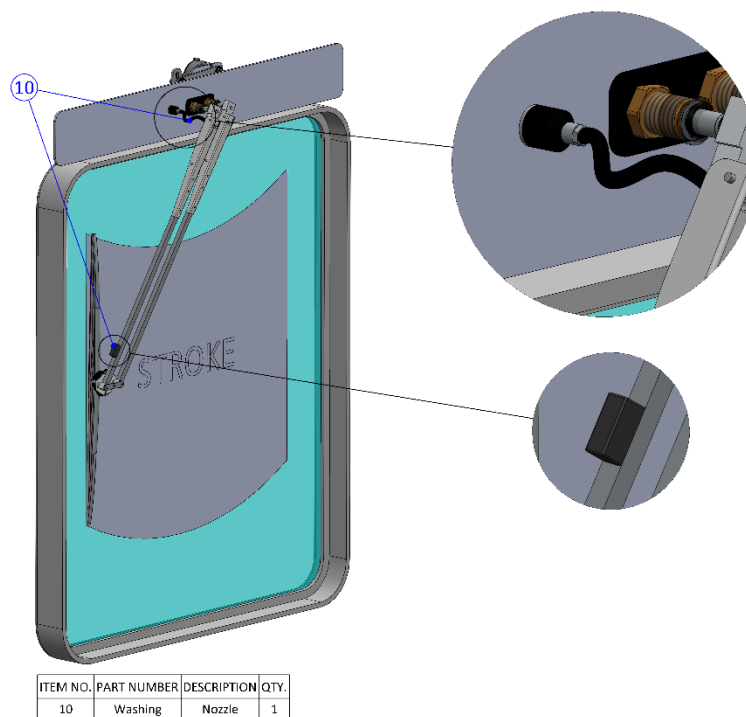


Figure 10. Washing Element

The installation personnel must provide the pressurized water source and the intermediate connection piping. When the washing option is installed, the maximum pressure for the system is 8 bar (118 PSI), and the minimum pressure for sufficient spray reach is 1 bar (15 PSI) (Table 4). Example flow rates for a single spray jet are shown below.

Pressure		Flow Rate	
Bar	Psi	Litres/min	Gallons/min
1.0	15	0.95	0.20
1.5	22	1.20	0.25
2.0	29	1.40	0.30
3.0	44	1.75	0.40

Tablo 4. Water System Pressure and Flow Rate

4. WIPER INSTALLATIONS

Attention: Confirm via the instruction form that the appropriate wiper, blade, and arms have been selected for each window.

Attention: Before starting the installation, check that there are no obstacles or hazards at the chosen location. The main frame must not obstruct the proper operation of the wiper, and the hull must provide a flat surface resistant to bending for mounting.

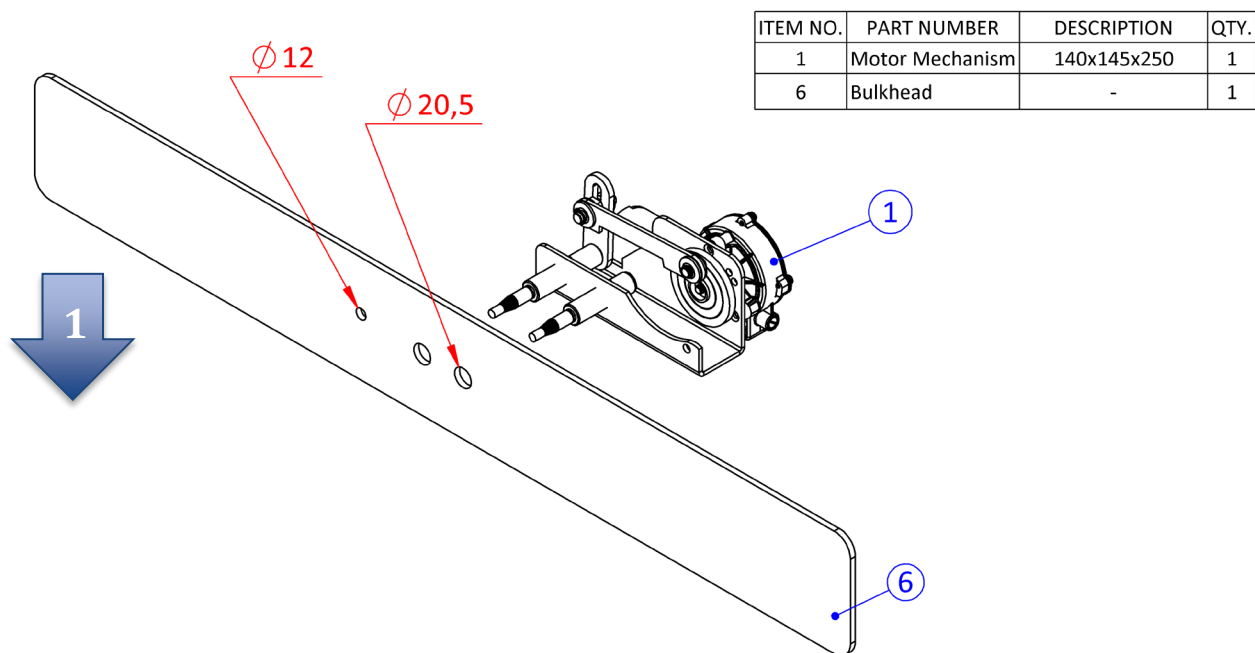
Attention: The motor mechanism must not come into contact with any supports inside the ship or the main ship plate.

On the ship's plate:

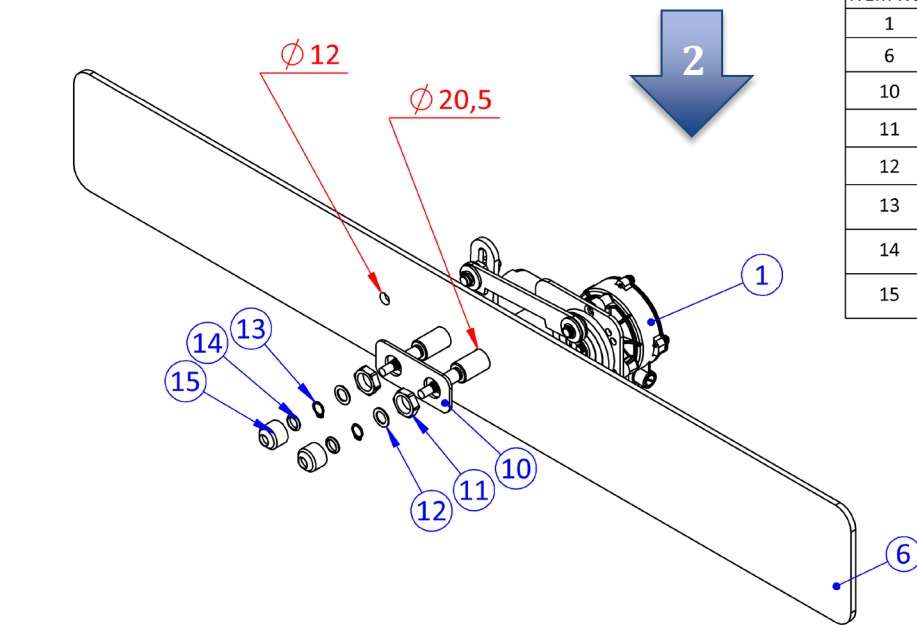
Drill a Ø20.5 mm hole for the motor shaft, ensuring the hole is circular and burrs are carefully removed. Treat the bare metal surface to prevent corrosion.

Attention: For pendulum type, drill 1 hole; for pantograph type, drill 2 holes. For the washing apparatus, drill 1 hole.

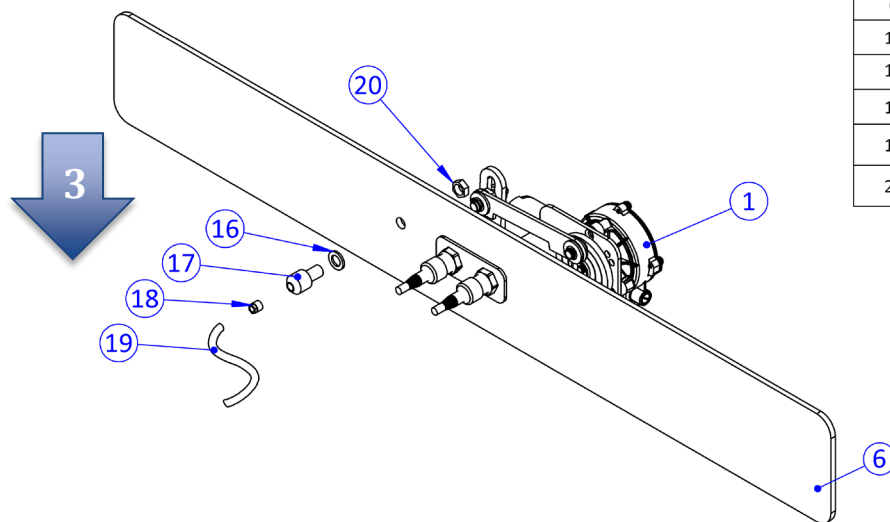
Leave sufficient spare cable length to allow removal of the motor assembly during maintenance.



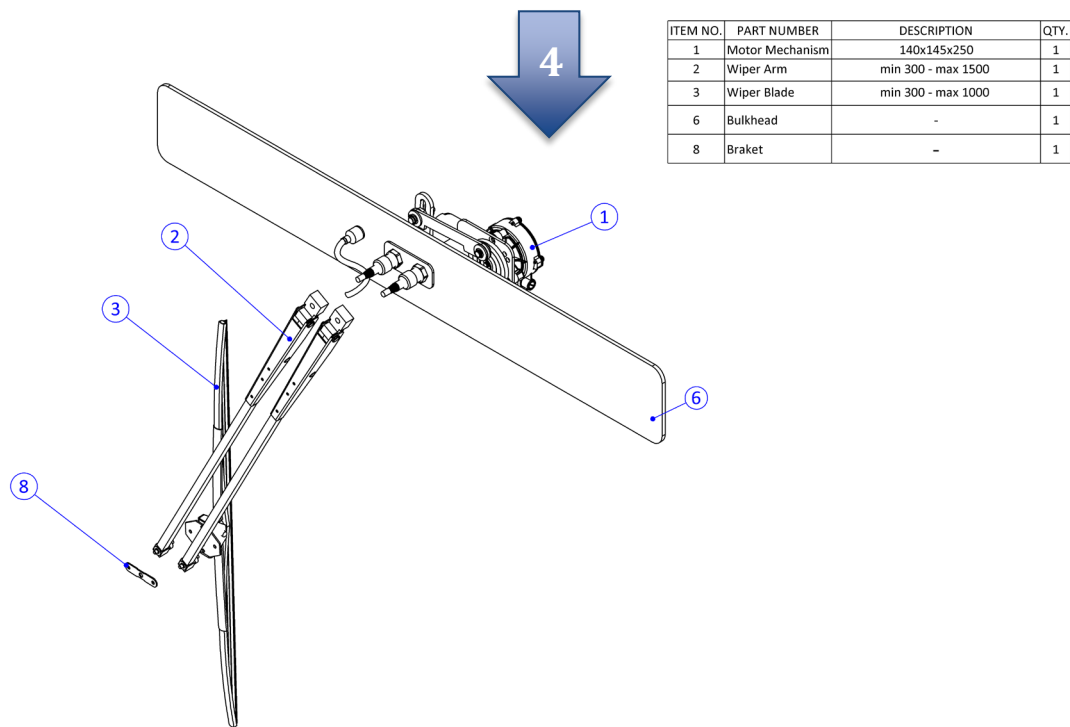
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ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Motor Mechanism	140x145x250	1
6	Bulkhead	-	1
10	Mil Braketi	3	1
11	Somun	M18	2
12	Pul	M18	2
13	Segman	15	2
14	O Ring	12	2
15	Cap	-	2



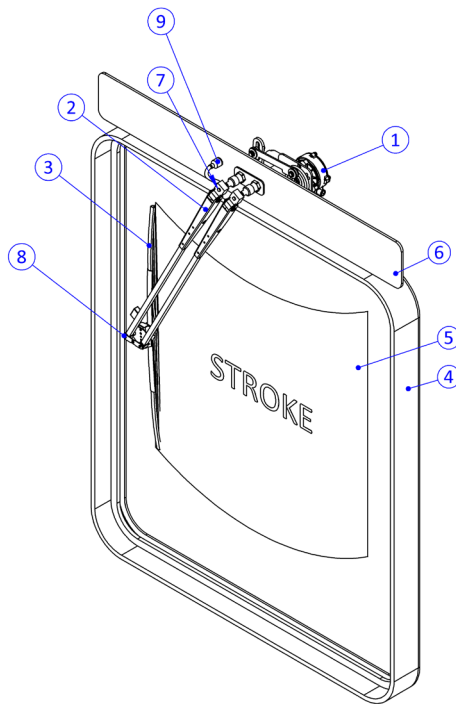
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Motor Mechanism	140x145x250	1
6	Bulkhead	-	1
16	Pul	M12	1
17	Saplama	M12	1
18	Somun	M12	1
19	Hortum	8	1
20	Somun	M12	1



1. The ship plate (6) is drilled with M20 and M12 diameter drills. The motor shafts (1) are passed through these holes.
2. After the motor shafts are inserted, the following parts are installed and secured in order: shaft bracket (10), M18 nut (11), M18 washer (12), circlip (13), O-ring (14), and cap (15).
3. For the washing system, the stud (17) is passed through the drilled M12 hole with a washer (16) placed in front. It is tightened and secured from the opposite side of the ship plate using a nut (20). The hose (19) is passed through the nut (18) and connected to the stud (17).
4. The wiper arm (2) is mounted onto the motor shaft and secured tightly with nuts. The wiper arm (2) is attached to the holes on the blade (3), and the bracket (8) is installed to complete the wiper assembly.

Move the wiper arm assembly through its full stroke and ensure the movement is unrestricted (the motor will provide some resistance, but it should not bind the wiper). Inspect for any restrictions and resolve them.

5. PART LIST



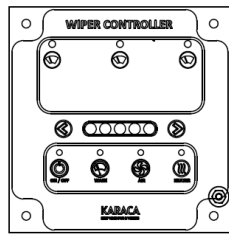
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Motor Mechanism	140x145x250	1
2	Wiper Arm	min 300 - max 1500	1
3	Wiper Blade	min 300 - max 1000	1
4	Window	WxH	1
5	Stroke	Depends on the wiper blade and arm	1
6	Bulkhead	-	1
7	Nut	M18	1
8	Support	316L	1
9	Washing	Nozzle	1

Figure 11. Part List

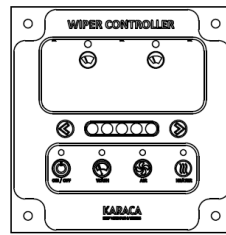
6. OLD – NEW CONTROL UNIT

6.1. GENERAL

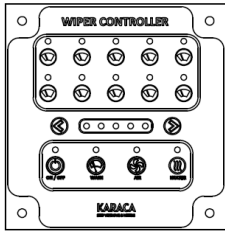
CT2N is a microcontroller-based wiper control system for up to 5 wiper motors; it provides all the functions for a proper windscreen cleaning. CT2N is composed by one (or more) control panels CP4PX (where 'X' stands for the number of wipers handled) and one relay box SRM2/3P or SRM4/5P depending on the number of wipers. Control panel is connected to relay box by one serial cable (CAN BUS CABLE) with two 8 pole plugs. Every unit has two plugs (in parallel) and any one or both can be used.



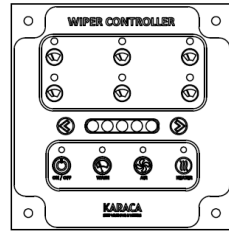
KRC-CP2-3way



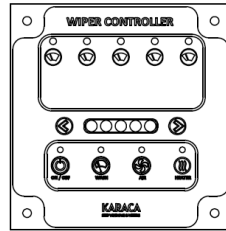
KRC-CP2-2way



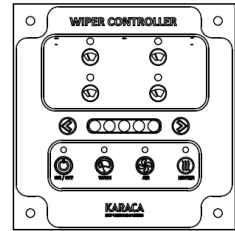
KRC-CP2-10way



KRC-CP2-6way



KRC-CP2-5way



KRC-CP2-4way

Features:

- ✓Double voltage supply: 12V- 24V
- ✓Activates/deactivates each wiper one by one
- ✓Synchronization in each selected mode
- ✓Three intermittent settings
- ✓Continuous slow and fast mode
- ✓Perfect self parking position due to "dynamic brake"
- ✓Wipe/wash program
- ✓Activating the heater
- ✓Easy to install and to use
- ✓Reliable

Standard supply includes:

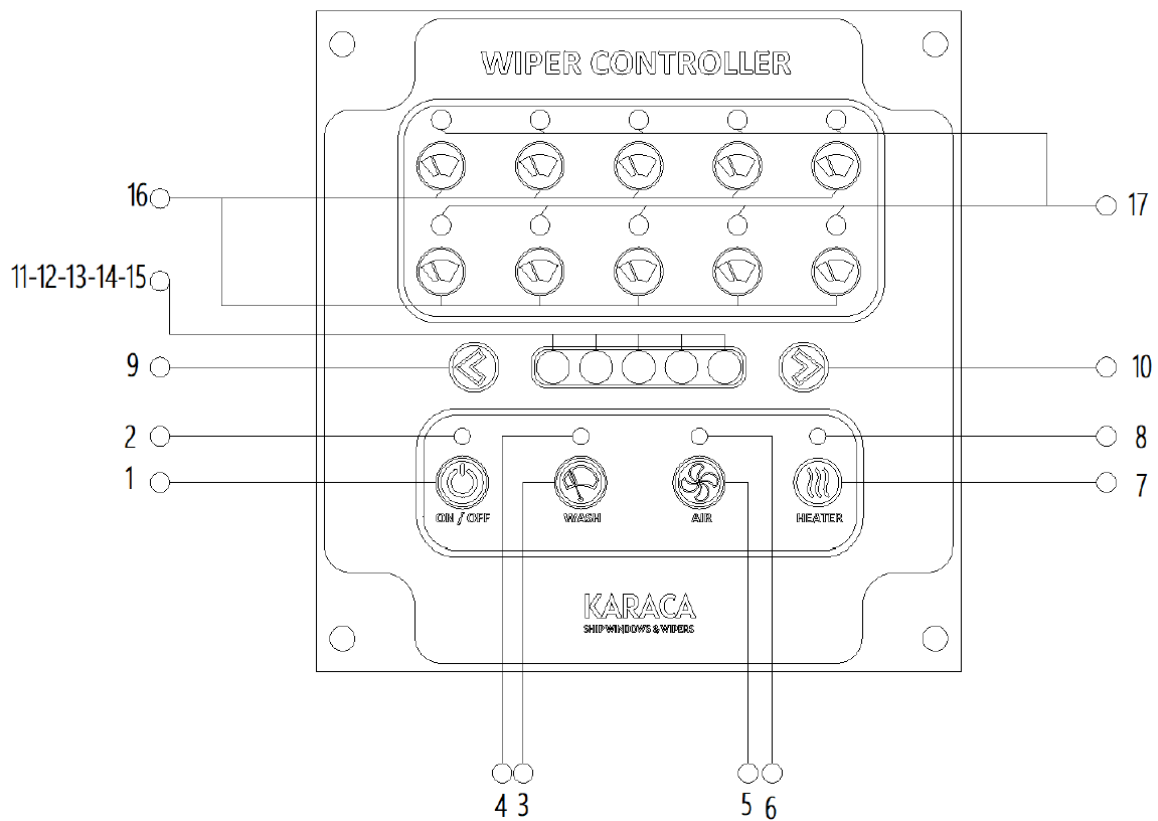
- ✓One KRCWCPX
- ✓One KRCWB5

✓Three solenoid valve

✓One cable for communication (standard 5 meters, armored cat 6 cable)

✓Female connectors for motors connection to KRCWB5(one for each motor)

6.2. PUSH BUTTON FUNCTIONS AND SIGNALLINGS



With reference to fig 1:

1. General ON / OFF button
2. General ON / OFF and fault led
3. Washing system ON / OFF button
4. Washing system ON / OFF led
5. Air system ON / OFF button
6. Air system ON / OFF led
7. Heater ON / OFF button
8. Heater ON / OFF led

9-10. Decrease speed / increase delay in intermittent mode

11-12-13-14-15. Selected mode led bar

16. Single wiper ON / OFF button

17. Single wiper ON / OFF and fault led

6.3. OPERATING MODE

Available operating mode.

SPEED BAR	WAITING TIME	MODE
ALL LEDS OFF	-	STAND-BY
LED 11 ON	6 SECONDS	INTERMITTENT
LED 11+12 ON	4 SECONDS	INTERMITTENT
LED 11+12+13 ON	2 SECONDS	INTERMITTENT
LED 11+12+13+14 ON	-	CONTINUOUS LOW SPEED
LED 11+12+13+14+15 ON	-	CONTINUOUS HIGH SPEED

6.4. WIPE / WASH / AIR PROGRAM

By pressing the wash button, washing program will start.

| 3 sec. | basılı tut | 4 sec. | 2 sec. |

|=== Spray ===== Spray =====|

|===== Wiper low speed or high speed=====|

|== Air ==|

*Press and hold the wash button to increase the washing time and spray/wipe time. When you release the wash button, automatic air will be activated. After the wipe/wash/air period the wipers will return to the selected park position.

6.5. FAILURES AND SIGNALLING

Wiper fault A flashing of a wiper led indicates that KRCWB5 detects an error on a wiper.

This may happen if:

- ✓ The parking switch is unconnected or damaged
- ✓ The motor doesn't run (internal self-resetting fuse in protection, motor unconnected or damaged)
- ✓ The motor doesn't run in FAST or in SLOW speed
- ✓ The motor can't be synchronized
- ✓ The motor connectors have been inserted after supplying power.

6.6. POWER SUPPLY

KRCWB5 will operate correctly if powered from 10V to 30V DC. Insert motor connectors BEFORE applying power supply. When powered, KRCWB5 will execute a lamp test and then will go in stand-by mode.

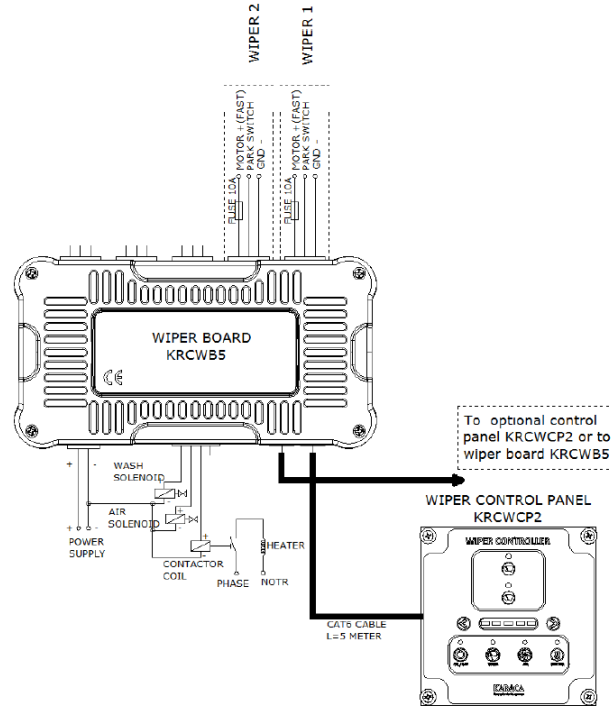
6.7. SYSTEM CONFIGURATION

KRCWB5 board can handle a maximum of 5 wipers. When the number of wipers is above 5, the number of KRCWB5 boards will also increase. The following configurations are available.

No. Of Installed Wipers	Wiper Control Panel	Wiper Board
2	KRCWCP2	KRCWB5
3	KRCWCP3	KRCWB5
4	KRCWCP4	KRCWB5
5	KRCWCP5	KRCWB5
6	KRCWCP6	KRCWB5X2
7	KRCWCP10	KRCWB5X2
8	KRCWCP10	KRCWB5X2
9	KRCWCP10	KRCWB5X2
10	KRCWCP10	KRCWB5X2

*User can connect 2 or more KRCWCPX and KRCWBX according to wiper groups using RJ45 sockets. The recommended connection method is to connect one control panel to the wiper board and not to the other control panel.

6.8. SYSTEM WIRING



7. MAINTENANCE AND CLEANING

External Surface Cleaning: Wipe the device's exterior with a damp cloth. Do not use abrasive cleaning materials.

Periodic Maintenance: The device's mechanical and electronic components should be checked every 6 months.

Wiper Blade Replacement: The wiper arm and blade can be replaced using the details of the Wiper Blade Assembly and Wiper Arm Assembly shown in Figure 13.